Numerical Reproduction of Particle Orbits in the Melvin Magnetic Universe

1. Introduction and Objectives

This report presents a numerical reproduction of selected results from the paper *Hypocycloid Motion in the Melvin Magnetic Universe*. The paper investigates the motion of charged test particles in the Melvin magnetic universe, focusing on hypocycloid-like trajectories and perturbed circular orbits.

The objectives of this work are to understand how the equations of motion and orbital constraints are implemented numerically, and to reproduce representative trajectories under weak- and strong-field conditions using Python.

Two cases are considered:

1. Hypocycloid-like motion in the weak-field regime.
2. Perturbed circular orbits in the strong-field regime.

2. Numerical Methods

2.1 Weak-Field Orbit

The weak-field case follows the equations of motion and orbital conditions discussed in Section 4 of the paper. The radial coordinate $r(\tau)$ and azimuthal coordinate $\phi(\tau)$ are obtained by numerically integrating the equations of motion.

The metric function is

$\Lambda(r) = 1 + \frac{B^2 r^2}{4}$

For the selected cusp-orbit conditions, the energy and angular momentum parameters are determined by

$E = \Lambda_*$

$L = \frac{eBr_*^2}{2\Lambda_*}$

where $\Lambda_* = \Lambda(r_*)$ and r_* is the reference cusp radius.

The trajectory is then transformed from cylindrical to Cartesian coordinates:

$x(\tau) = r(\tau)\cos\phi(\tau)$

$y(\tau) = r(\tau)\sin\phi(\tau)$

The resulting orbit is visualised using Python.

2.2 Strong-Field Perturbed Circular Orbit

The strong-field case follows the first-order perturbation analysis in Section 3 of the paper. The radial coordinate is approximated by

$$r(\tau) = r_0 + \varepsilon \cos(\omega\tau)$$

The corresponding first-order expression for the azimuthal coordinate is

$$\phi(\tau) = \frac{\beta_0}{\omega}(\omega\tau - \zeta \sin(\omega\tau))$$

Here, r_0 is the radius of the unperturbed circular orbit, ε is the perturbation amplitude, ω is the radial oscillation frequency, and β_0 is the angular frequency of the unperturbed orbit. The parameter ζ characterises the local cycloidal structure.

Unlike the weak-field case, this calculation uses the paper's first-order perturbative expressions rather than directly integrating the full nonlinear equations of motion.

3. Weak-Field Hypocycloid-Like Orbit

3.1 Parameters

The following parameters were used:

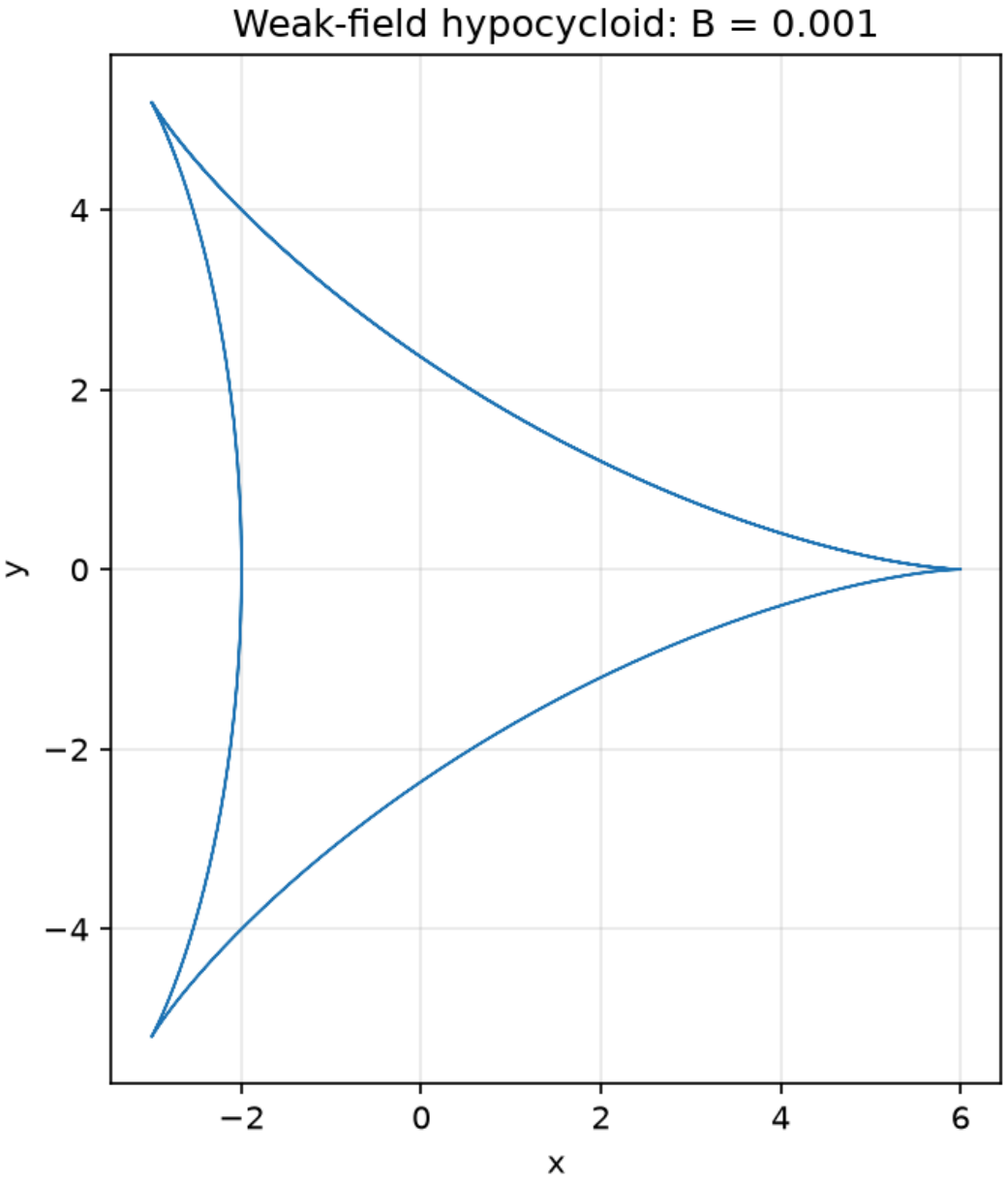
- Magnetic-field parameter: $B = 0.001$
- Charge parameter: $e = 0.5$
- Reference cusp radius: $r_* = 6$
- Axial momentum: $P = 0$

The calculated quantities are:

- $\Lambda_* = 1.0000090000$
- $E = 1.0000090000$
- $L = 0.0089999190$

3.2 Numerical Result

The numerical trajectory exhibits the expected three-cusped hypocycloid-like shape, qualitatively consistent with the weak-field behaviour discussed in the paper.



Suggested figure: The complete weak-field trajectory, with all three cusps visible and equal scaling on the two coordinate axes.

3.3 Discussion

This calculation provides a reproduction of the weak-field orbit and establishes a numerical workflow from parameter selection and integration of the equations of motion to trajectory visualisation.

A systematic analysis of numerical convergence and errors relative to an analytical trajectory has not yet been carried out.

4. Strong-Field Perturbed Circular Orbit

4.1 Parameters

The strong-field example uses the parameters associated with Figure 2(a) of the paper:

- Magnetic-field parameter: $B = 0.1$

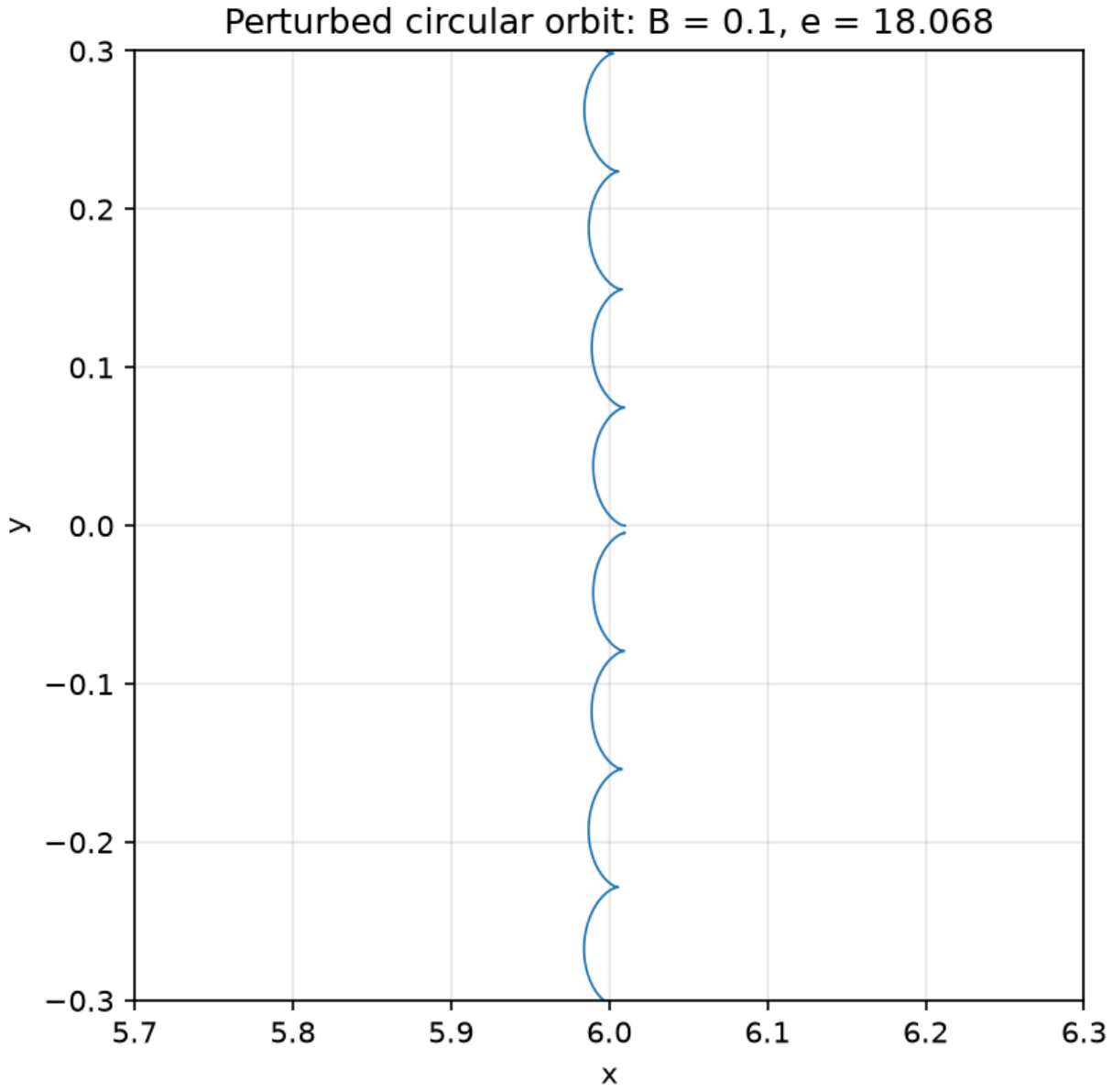
- Charge parameter: $e = 18.068$
- Radius of the unperturbed circular orbit: $r_0 = 6$
- Perturbation amplitude: $\varepsilon = 0.01$

The program produced the following values:

- Angular momentum parameter: $L = 29.9283600965$
- Squared energy parameter: $E^2 = 1.1884268179$
- Energy parameter: $E = 1.0901499062$
- Radial oscillation frequency: $\omega = 1.5247733493$
- Angular frequency: $\beta_0 = 0.0030130175$
- Frequency ratio: $\omega/|\beta_0| = 506.0619$
- Cycloidal shape parameter: $\zeta = 1.0095991712$

4.2 Numerical Result

The radial coordinate undergoes a small oscillation around $r_0 = 6$ with amplitude $\varepsilon = 0.01$. Consequently, the perturbed trajectory lies close to the unperturbed circular orbit, and its local structure is best examined using an enlarged view.



Suggested figure: The enlarged trajectory near the unperturbed circular orbit, showing the local perturbation structure.

4.3 Comparison with the Paper

The paper reports a frequency ratio of approximately

$$\frac{\omega}{\beta_0} \simeq 506.1$$

The present calculation gives

$$\frac{\omega}{|\beta_0|} = 506.0619$$

These values agree to the reported precision, providing a quantitative consistency check for the frequency ratio.

However, the calculated shape parameter $\zeta = 1.0095991712$ differs from the related value of approximately 1.0903 reported in the caption of Figure 2(a). This discrepancy has not yet been resolved. The expression for the angular expansion coefficient, the angular-momentum

branch, and the relationship between the symbols used in the text and figure caption require further verification.

Therefore, the present strong-field result should be regarded as a reproduction of the perturbative orbit structure rather than an exact quantitative reproduction of every detail in Figure 2(a).

5. Summary of Results and Limitations

5.1 Results Obtained

1. A Python workflow has been established for calculating and visualising particle trajectories in the Melvin magnetic universe.
2. The weak-field calculation produces a hypocycloid-like orbit with the expected qualitative shape.
3. The strong-field calculation produces a local perturbed-orbit structure using the first-order perturbation formulas.
4. The calculated frequency ratio $\omega/|\beta_0| = 506.0619$ agrees with the value of approximately 506.1 reported in the paper.
5. The relationship between the orbital parameters and the resulting trajectory has been explored through numerical implementation.

5.2 Remaining Limitations

1. The strong-field calculation uses a first-order perturbative approximation rather than direct integration of the full nonlinear equations of motion.
2. The discrepancy between the calculated ζ and the value reported in the figure caption remains unresolved.
3. Numerical convergence and approximation errors have not yet been systematically assessed.
4. An exact analytical solution or a complete set of conditions for the corresponding hypocycloid structure has not yet been established.

6. Conclusion and Future Work

This work provides a preliminary numerical reproduction of two orbital regimes discussed in the paper: hypocycloid-like motion in the weak-field regime and perturbed circular motion in the strong-field regime.

The weak-field trajectory is qualitatively consistent with the expected hypocycloid-like structure. In the strong-field case, the calculated frequency ratio agrees with the value reported in the paper. However, a discrepancy remains in the cycloidal shape parameter, and further verification is needed before claiming a complete quantitative reproduction.

The next step is to investigate the conditions under which the cycloidal structure admits an exact solution, verify the relevant parameter relations, and compare any resulting analytical trajectory with the numerical calculations.